

CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE



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USER STORY

The **Customer Support Ticket Priority Prediction and Automated Assignment System** is a Salesforce-based solution that uses **Agentforce and Salesforce Flow** to automatically analyze customer support tickets, evaluate ticket descriptions and customer sentiment, and classify each ticket as **High, Medium, or Low priority**. The system is designed to reduce manual ticket review and help support teams identify and handle critical issues more efficiently.

The solution uses the **Support_Ticket_Intelligence__c** custom object to store ticket and customer information, an **Auto-Launched Flow** to perform backend processing, and an Agentforce subagent named **Support Ticket Priority Analysis** to provide conversational access to the ticket-analysis process. Agentforce retrieves the relevant customer ticket and triggers the Flow to determine its priority and appropriate support assignment.

For critical cases, particularly **High-priority or negative-sentiment tickets**, the system can automatically create an urgent handling task, route the case to a **Tier 2 Technical Specialist**, and check for potential **SLA breach risk**. Lower-priority requests can be routed to the **General Support Queue** without unnecessary escalation.

Overall, the system combines **AI-assisted analysis and Salesforce automation** to improve ticket prioritization, reduce administrative effort, support consistent assignment, and enable faster response to critical customer issues.

1. INTRODUCTION

1.1 Project Overview :

Customer support teams handle a large volume of tickets every day, making manual prioritization and assignment time-consuming and sometimes inconsistent. This project provides a Salesforce-based solution that automatically analyzes ticket descriptions and customer sentiment, classifies ticket urgency as High, Medium, or Low, assigns the appropriate support level, and triggers backend automation for critical cases. The solution helps support teams identify urgent issues faster, reduce manual effort, and improve overall ticket-handling efficiency.

1.2 Purpose

- Automatically classify support tickets as High, Medium, or Low priority based on issue description and sentiment analysis.
- Retrieve the latest support ticket associated with a customer Account.
- Reduce manual prioritization and assignment effort.
- Create an urgent handling task on the customer Account for High-priority tickets.
- Use Agentforce to analyze ticket details conversationally and trigger backend automation.
- Prevent service failures through automated SLA breach risk detection and database record flagging.
- Improve ticket response and resolution efficiency.

2. IDEATION PHASE

2.1 Problem Statement

PS ID	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Support Agent	focus on critical tickets first	tickets are not always prioritized automatically	ticket prioritization can be manual	frustrated
PS-2	Customer	get urgent issues resolved quickly	critical issues may wait behind lower-priority tickets	support teams handle large ticket volumes	concerned
PS-3	Support Manager	assign tickets to	assignment may require	ticket urgency is	overwhelmed

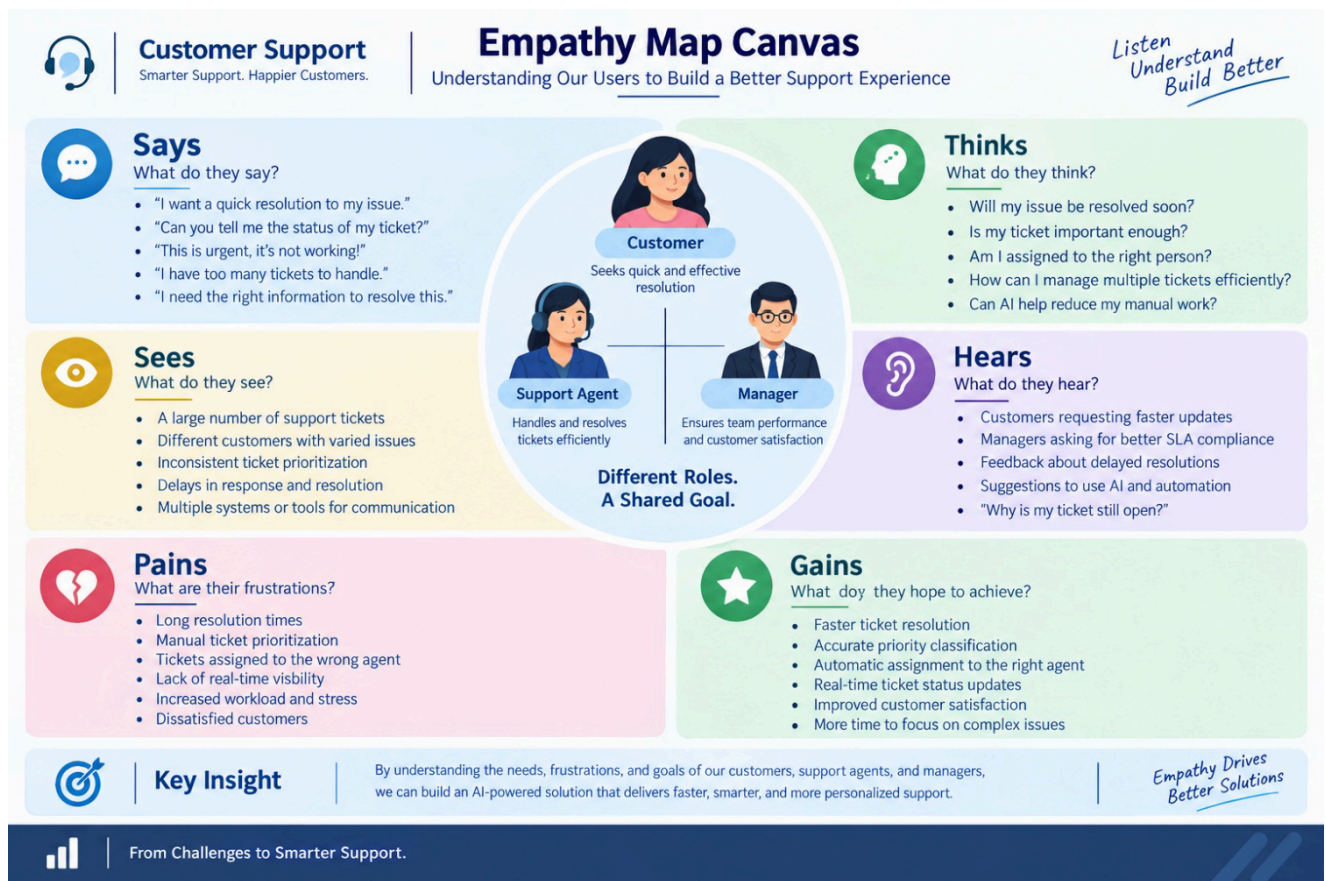
		the right support level	manual decisions	not consistently analyzed	
PS-4	Support Agent	identify urgent issues from descriptions	important keywords can be missed	ticket descriptions contain different urgency signals	uncertain
PS-5	Support Manager	reduce SLA-related risk	older unresolved tickets may remain unnoticed	SLA risk needs additional checking	concerned



2.2 Empathy Map Canvas

User Persona: Support Agent

- **Says:** "I want a quick resolution to my issue," "This is urgent, it's not working!", "I have too many tickets to handle," "I need the right information to resolve this".
- **Thinks:** Will my issue be resolved soon? Is my ticket important enough? Am I assigned to the right person? Can AI help reduce my manual work?.
- **Hears:** Customers requesting faster updates, managers asking for better SLA compliance, feedback about delayed resolutions.
- **Sees:** A large number of support tickets, different customers with varied issues, inconsistent ticket prioritization, and delays in response.
- **Pains:** Long resolution times, manual ticket prioritization, tickets assigned to the wrong agent, lack of real-time visibility, increased workload, and dissatisfied customers.
- **Gains:** Faster ticket resolution, accurate priority classification, automatic assignment to the right agent, real-time ticket status updates, and more time to focus on complex issues.



2.2 Brainstorming & Idea Prioritization

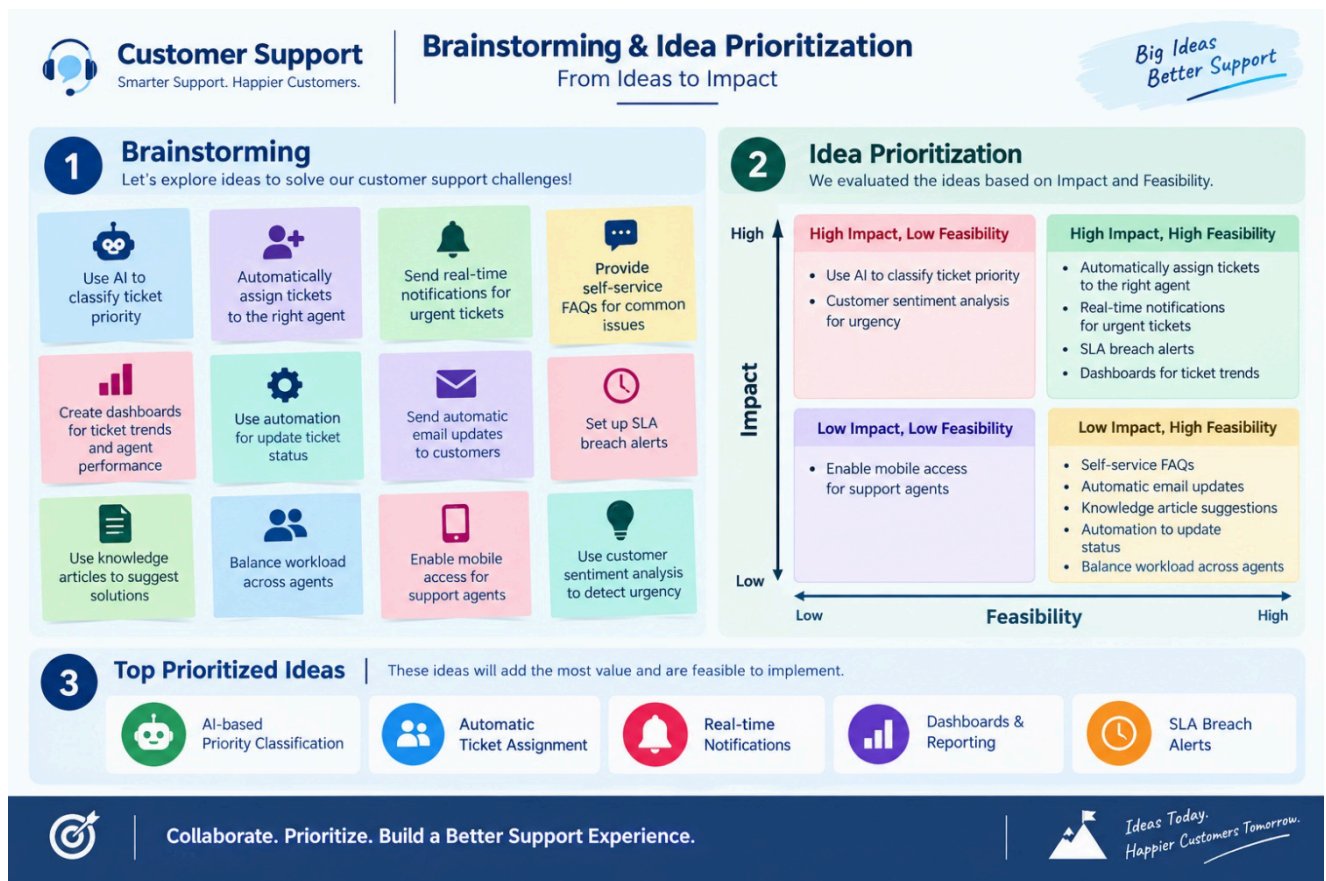
Ideas Considered:

- Manual ticket registers
- Spreadsheet-based ticket tracking
- Standalone support portal
- Salesforce CRM with Agentforce

Final Selection:

Salesforce CRM selected due to:

- Automation through Salesforce Flow
- AI-assisted ticket analysis through Agentforce
- Automatic priority classification
- Automatic urgent-task creation
- Reduced manual support effort.



3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

Ticket Creation → Account Identification → Latest Support Ticket Retrieval → Ticket Description and Sentiment Analysis → Priority Classification (High/Medium/Low) → Support Level Assignment → Urgent Task Creation for Critical Tickets → SLA Risk Verification → Support Handling → Ticket Resolution and Status Update.



3.2 Solution Requirements

Functional Requirements :

The functional requirements define the key functions that the Customer Support Ticket Priority Prediction and Automated Assignment System must perform. These include ticket management, ticket analysis, priority classification, support assignment, urgent task creation, SLA risk checking, and Agentforce-based interaction.

FR No	Requirement
FR-1	Support ticket creation and management
FR-2	Account and Contact association
FR-3	Ticket description analysis
FR-4	Automatic priority classification
FR-5	High-priority task creation
FR No	Requirement
FR-6	Support agent assignment
FR-7	SLA breach risk checking
FR-8	Agentforce conversational analysis

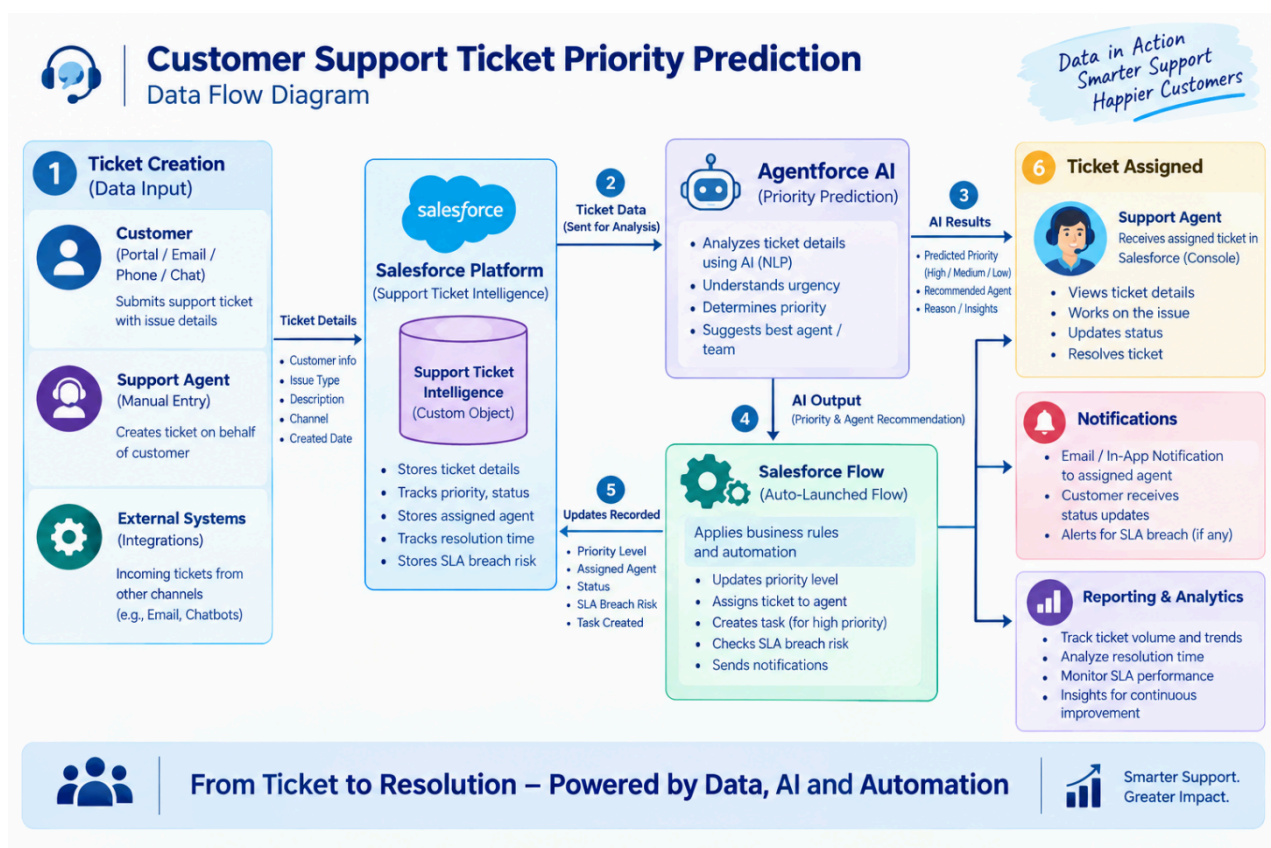
Non-Functional Requirements

NFR	Description
Usability	Clear Agentforce interaction and understandable priority messages.
Reliability	Consistent priority classification based on configured conditions.
Performance	Quick retrieval and analysis of the latest support ticket.
Security	Controlled Salesforce access to customer and ticket information.
Maintainability	Flow and Agentforce configuration can be updated.
Scalability	Supports increasing support-ticket volumes through Salesforce automation.

3.3 Data Flow Diagram

- Flow Execution Path: Customer / Support Request → Account → Latest Support Ticket → TicketDescription → Analyze Description → Priority Level (High/Medium/Low) Decision → Agent Assignment → High-Priority Task (if applicable) → Final Response.

- Agentforce Invocation Path: User provides Account Name → Agentforce Support Ticket Priority Analysis → Auto-Launched Flow → Ticket Analysis → Priority and Assignment Output → Response to User.
- The user provides the Account Name to the Agentforce **Support Ticket Priority Analysis** subagent. Agentforce invokes the **Auto-Launched Flow**, which identifies the associated Account and retrieves the latest support ticket. The Flow analyzes the ticket description and sentiment, determines the appropriate priority level, assigns the required support level, and triggers urgent task creation when necessary. The final priority, assignment, and action details are then returned to the user through the Agentforce response.



3.4 Technology Stack

Layer	Technology
Platform	Salesforce

Database	Support_Ticket_Intelligence__c with Account, Contact, User and Task records
Logic	Auto-Launched Flow
AI	Agentforce
Agent Configuration	Support Ticket Priority Analysis subagent
Automation	Flow Decision, Assignment and Create Records
Output	Agentforce conversation response

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem	Solution	Benefit
Manual ticket prioritization	Flow analyzes description and sets priority	Faster identification of urgent cases
Urgent issues may be missed	High-priority keyword decision	Critical cases receive attention
Manual task creation	Flow creates Task for High priority	Reduced manual effort
Problem	Solution	Benefit
Manual support assignment	Flow sets assigned support level	More consistent handling
SLA risk can be overlooked	Optional SLA Risk decision	Better risk awareness
Need for conversational access	Agentforce subagent	AI-assisted interaction



4.2 Proposed Solution

Problem: Manual ticket prioritization and assignment causing delays in handling critical customer support issues.

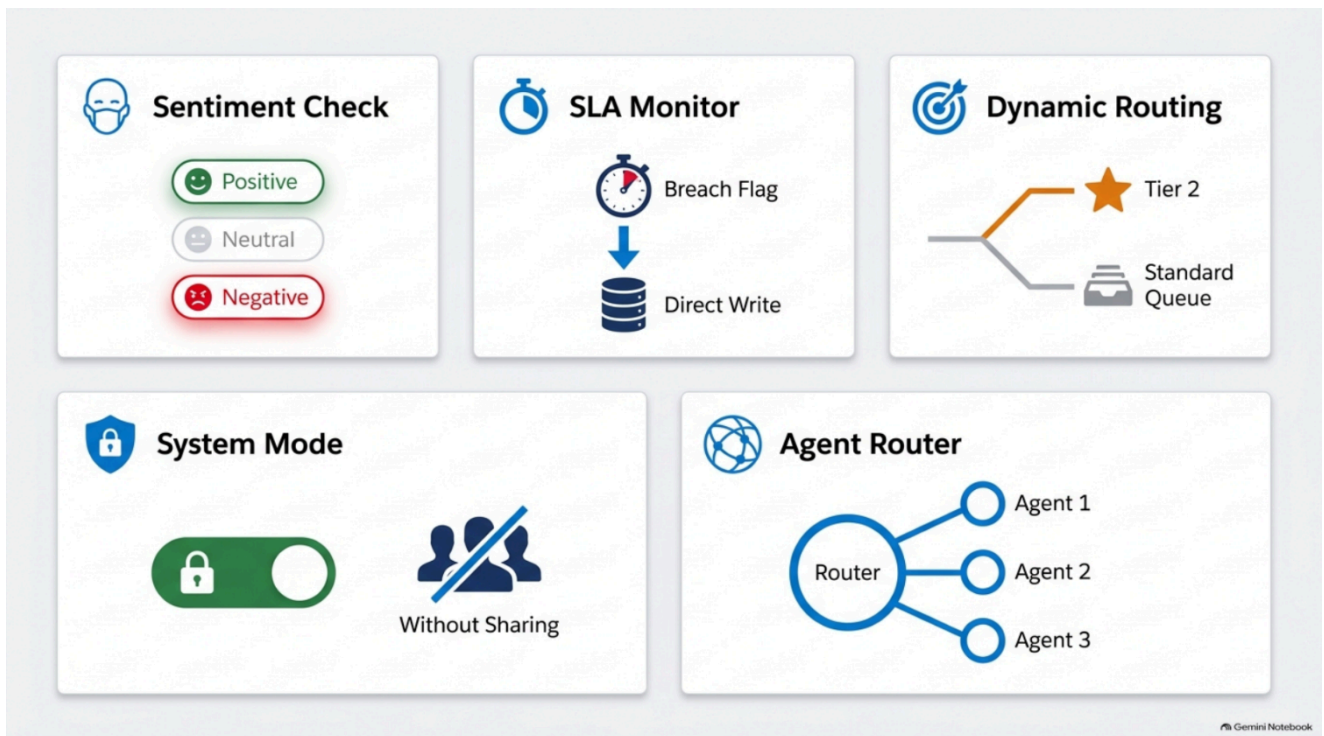
Solution: Salesforce-based Customer Support Ticket Priority Prediction and Automated Assignment System.

Innovation: Agentforce AI + Auto-Launched Flow for analyzing ticket descriptions, automatically classifying priority, assigning support level, and triggering urgent-task automation.

Impact: Faster identification of critical tickets, reduced manual effort, improved response time, and better support-team productivity.

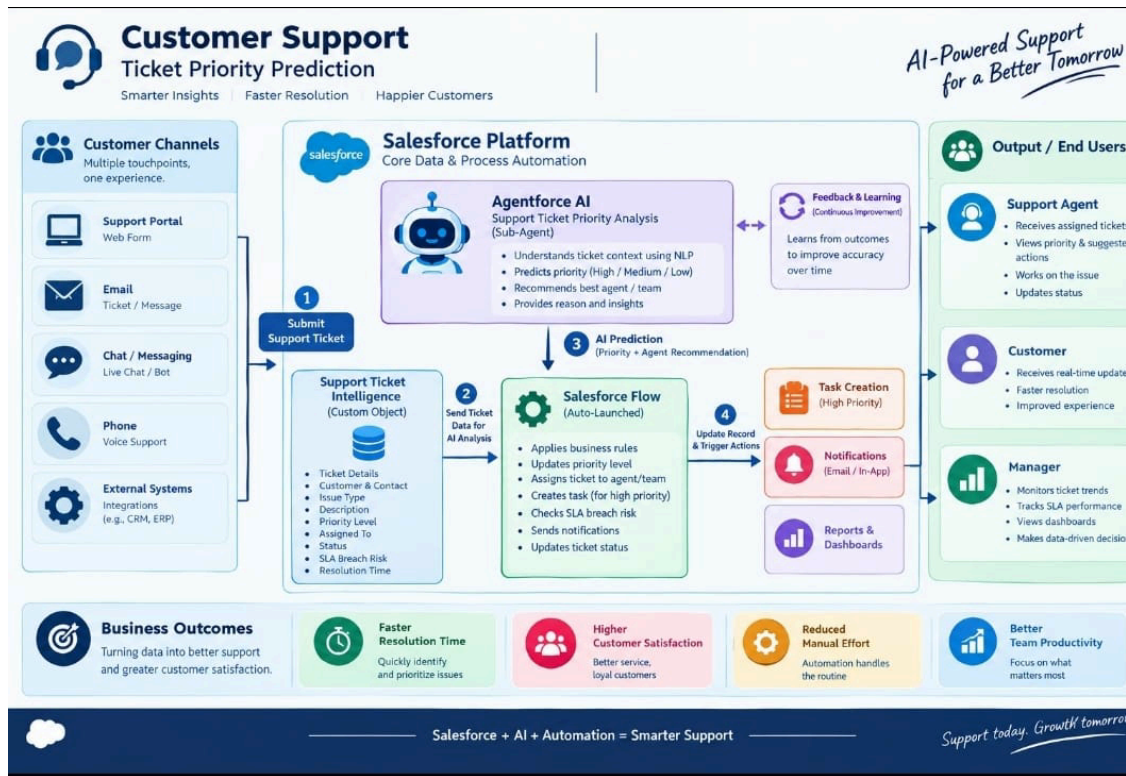
Business Model: Customer support operations involving customers, support agents, support managers, and AI-assisted ticket handling.

Scalability: Supports increasing support-ticket volumes through Salesforce automation, with scope for additional assignment rules, SLA monitoring, and priority indicators.



4.3 Solution Architecture

Users / Support Team → Agentforce → Support Ticket Priority Analysis → Auto-Launched Flow → Account & Ticket Data → Description Analysis → Priority Decision → Task / Assignment → Final Message → Agentforce Response.



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Methodology: Agile framework with Sprints, User Stories, and Story Point estimations.

Product Backlog & Sprint Schedule

Sprint	Epic	USN	User Story	Points	Priority
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to configure the Salesforce environment for Support Ticket Intelligence...	3	High
Sprint-2	Data Modeling	USN-2	As an admin, I want to create the Support Ticket Intelligence object with required fields...	5	High
Sprint-2	Data Modeling	USN-3	As a support user, I want support tickets to be	3	High

			associated with Accounts and Contacts...		
Sprint-3	Automation	USN -4	As a support agent, I want an Auto-Launched Flow to retrieve the latest ticket and analyze its description...	5	High
Sprint-3	Automation	USN -5	As a support agent, I want tickets to be automatically classified as High, Medium, or Low...	5	High
Sprint-4	Agentforce	USN -6	As an AI Agent, I want to analyze support ticket details and trigger the backend Flow...	8	High
Sprint	Epic	USN	User Story	Points	Priority
Sprint-4	Task Assignment	USN -7	As a support manager, I want High-priority tickets to automatically create an urgent handling task...	5	High
Sprint-5	SLA Management	USN -8	As a support manager, I want the system to check SLA breach risk for older unresolved tickets...	5	Medium
Sprint-6	Conversational	USN -9	As a support user, I want to provide an Account Name to Agentforce and receive the ticket outcome...	5	Medium

5.2 Project Tracker

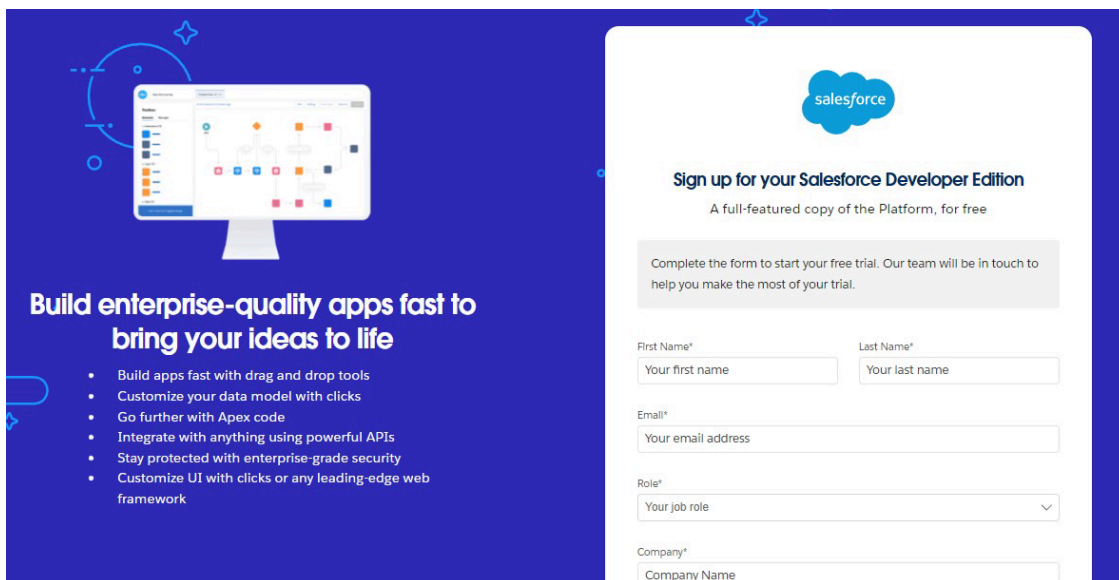
Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022

Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022

6. PROJECT DEVELOPMENT PHASE

Milestone 1: Salesforce Environment Setup

- Created a Salesforce Developer Edition environment.
- Activated org access and configured administrative privileges.



Milestone 2: Data Management – Custom Objects & Fields

Object Name: Support Ticket

Intelligence API Name:

Support_Ticket_Intelligence__c

Developer Edition

Search Setup

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Support Ticket Intelligence

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Fields & Relationships

15 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

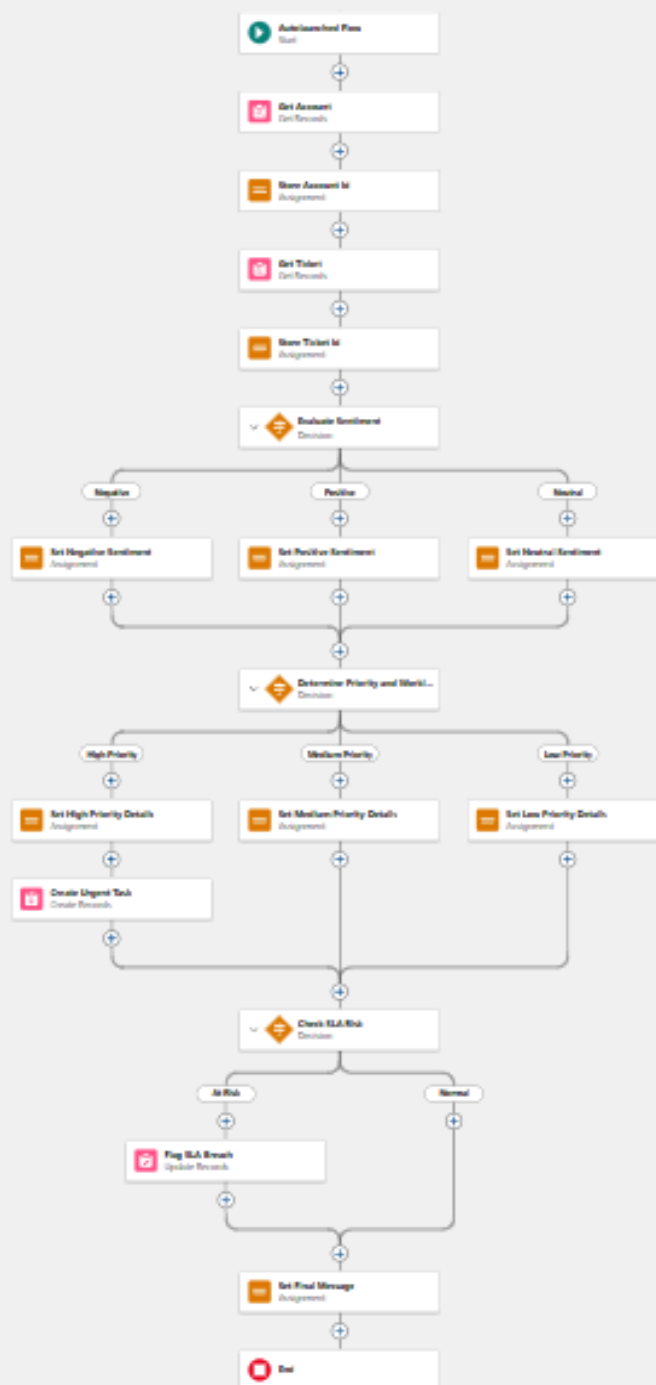
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Assigned Agent Role	Assigned_Agent_Role__c	Text(100)		
Contact	Contact__c	Lookup(Contact)		✓
Created By	CreatedById	Lookup(User)		
Customer	Customer__c	Lookup(Account)		✓
Description	Description__c	Long Text Area(32768)		
Issue Type	Issue_Type__c	Picklist		
Last Modified By	LastModifiedById	Lookup(User)		

Fields Created:

Field API Name	Type / Relationship	Purpose
Ticket_Number__c	Auto Number	Ticket number
Customer__c	Lookup Account	Customer Account
Contact__c	Lookup Contact	Customer contact
Issue_Type__c	Picklist	Issue classification
Description__c	Long Text Area	Issue details
Priority_Level__c	Picklist	Priority level
Status__c	Picklist	Ticket status
Created_Date__c	Date	Ticket date
Assigned_To__c	Lookup User	Support agent
SLA_Breach_Risk__c	Checkbox	SLA risk flag
Resolution_Time__c	Number	Time taken

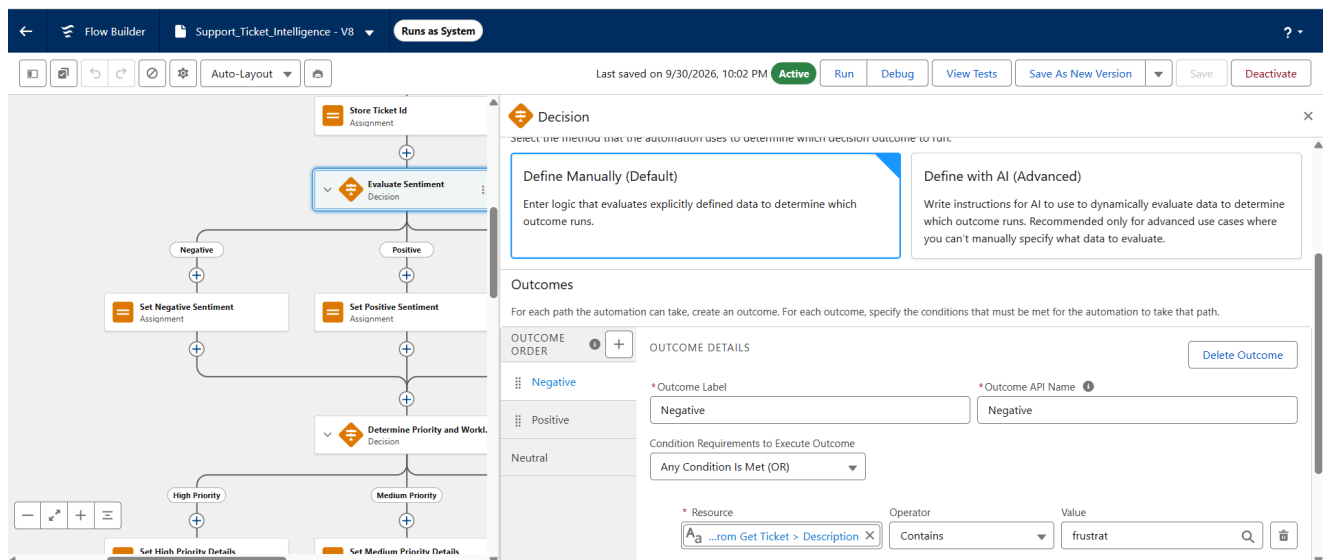
Milestone 3: Auto-Launched Flow Logic (Support_Ticket_Intelligence)

- Type: Auto-Launched Flow (No Trigger, called from Agentforce).
- Input Variables: varAccountName (Text).
- Output Variables: varAccountId, varTicketId, varPriorityLevel, varAssignedTo, varActionMessage.
- Logic Path: Start → Get Account → Store Account Id → Get Ticket → Store Ticket Id → Decision (AnalyzeDescription) → Assignment (Priority Level) → Decision (High Priority) → Create Task → Assignment (Assigned Agent) → Optional SLA Decision → Assignment (Final Message) → End.



Milestone 4: Agentforce Subagent Configuration

- Name: Support Ticket Priority Analysis.
- API Name: Support_Ticket_Priority_Analysis.
- Scope: Analyze support ticket descriptions, determine the priority level, and trigger backend automation for task assignment. Do not handle unrelated requests such as billing or account updates.
- Agent Action: Support_Ticket_Intelligence__c analyzes customer support tickets based on the provided account name and determines ticket priority.



7. ADVANCED VALUE ADDITIONS (BEYOND BASE BLUEPRINT)

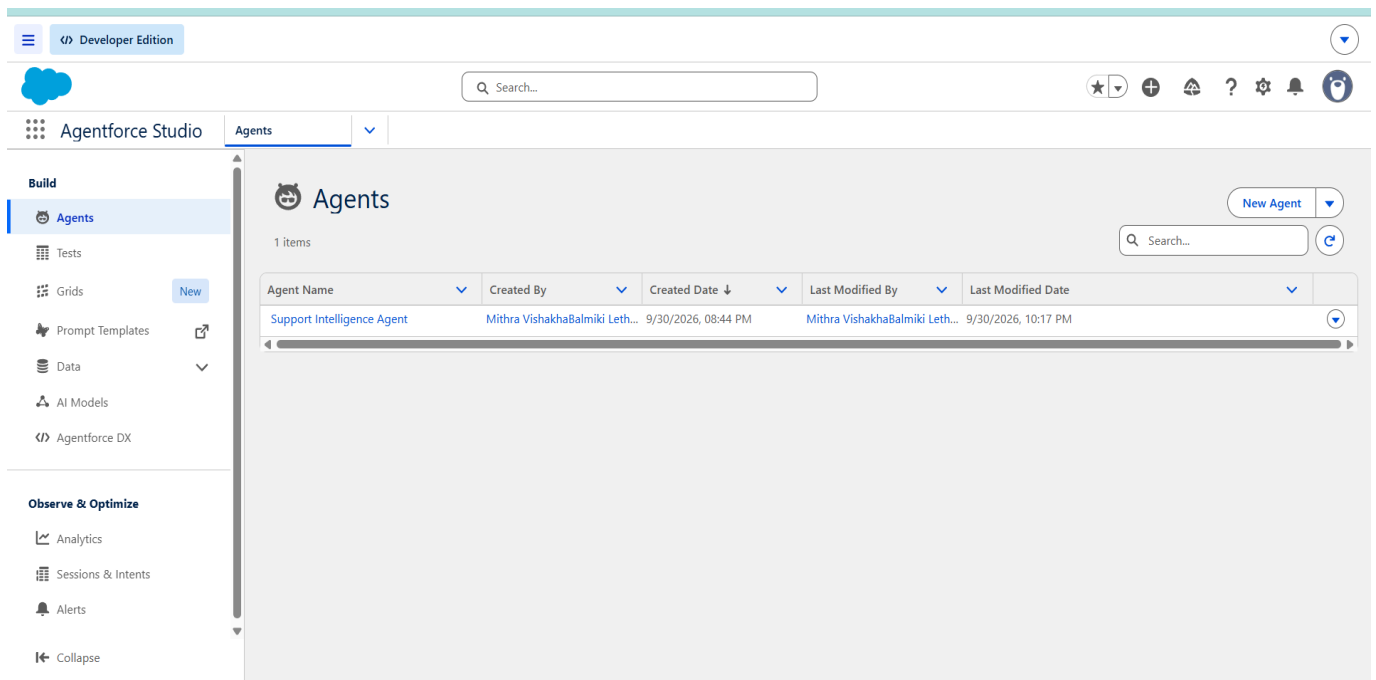
1. Multidimensional Sentiment Evaluation: Evaluates emotional sentiment (Negative, Neutral, Positive) alongside raw keyword syntax, enabling faster detection of sensitive escalations.
2. Automated SLA Breach Risk Flagging: Upgraded the SLA check from an optional read into an active update operation that dynamically writes `SLA_Breach_Risk__c = True` directly to the Salesforce record.
3. Dynamic Specialist Workload Routing: Replaced static text assignments with role-based routing (Tier 2 Technical Specialist for critical escalations vs. General Support Queue for routine inquiries).

4. System Mode Without Sharing Security Architecture: Configured the Flow to run in System Mode without Sharing, preventing permission exceptions for background Agentforce operations.
5. Enterprise Multi-Agent Canvas Orchestration: Subagent configured within the full Agentforce Builder architecture, managed by the central Agent Router.

8. FUNCTIONAL & PERFORMANCE TESTING

During testing, scenarios were executed in the Agentforce chat simulator and verified on live records in Salesforce:

Test Scenario	Input / Condition	Expected / Observed Outcome
Critical Incident	“Urgent: System failure preventing login”	High priority detected; Urgent Ticket Handling task created; SLA risk flagged; Tier 2 Specialist assigned.
Standard Request	“General inquiry regarding standard account features”	Low priority detected; task creation bypassed; routed to General Support Queue.



9. ADVANTAGES & DISADVANTAGES

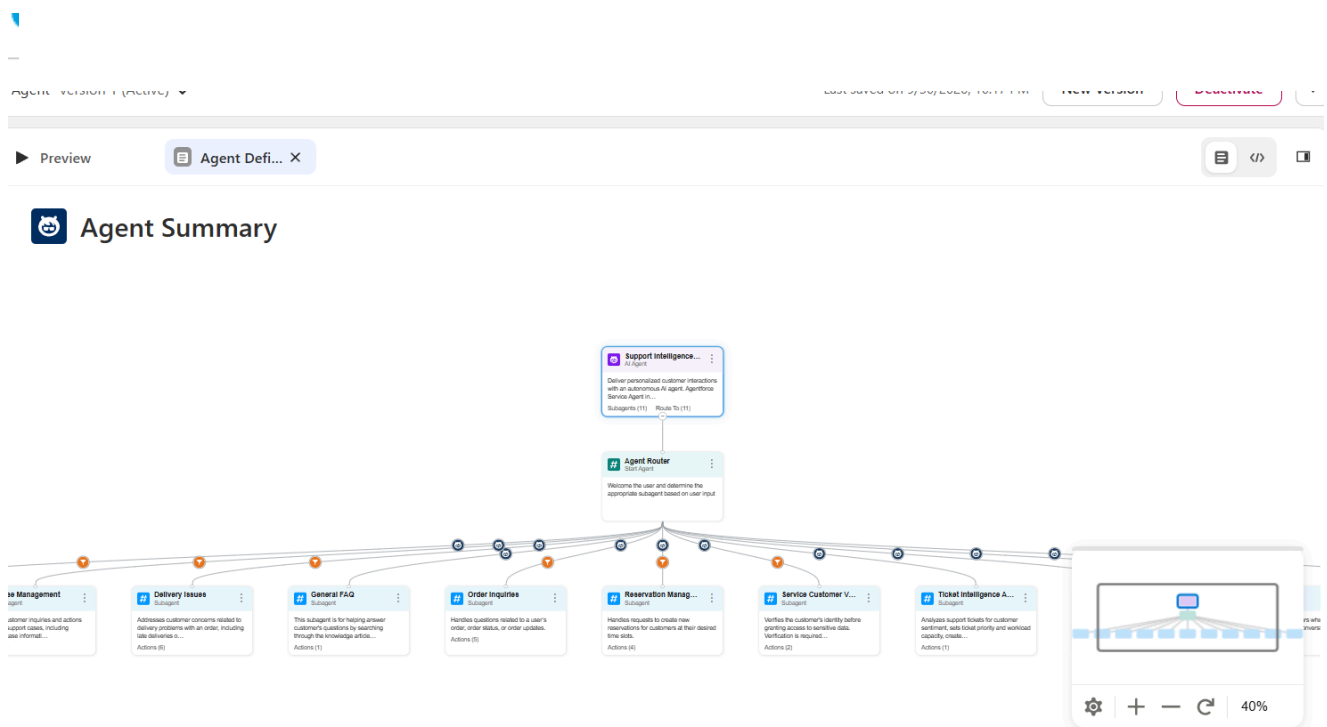
Advantages

- **Automatic Ticket Prioritization:** Automatically classifies support tickets as High, Medium, or Low based on the ticket description and configured priority rules.
- **Faster Identification of Critical Issues:** Helps support agents quickly identify urgent tickets and focus on cases that require immediate attention.
- **Reduced Manual Effort:** Automates ticket analysis, priority classification, support assignment, and related backend actions, reducing repetitive administrative work.
- **Automatic Urgent Task Creation:** High-priority tickets automatically generate an Urgent Ticket Handling task, ensuring critical cases receive prompt attention.
- **SLA Risk Awareness:** Automatically checks older unresolved tickets and flags potential SLA breach risks, helping support teams take corrective action in time.

- **Improved Support Efficiency:** Agentforce and Salesforce Flow work together to provide faster ticket analysis and more consistent support handling.

Disadvantages

- **Keyword-Based Priority Logic:** Priority classification depends on configured keywords and rules, which may not always capture the full context or complexity of a customer issue.
- **Dependence on Accurate Data:** The system requires accurate Account, Contact, and support-ticket information to retrieve records and perform analysis successfully.
- **Configuration Maintenance:** Flow logic, priority rules, and Agentforce subagent instructions need to be reviewed and updated as business requirements change.
- **Limited Context Understanding:** Complex or unusual customer issues may require additional human review when predefined rules do not fully represent the situation.



Developer Edition

Search...

Agentforce Studio

Agents

Build

Agents

Tests

Grids

Prompt Templates

Data

AI Models

Agentforce DX

Observe & Optimize

Analytics

Sessions & Intents

Alerts

Collapse

Agents

1 items

New Agent

Search...

Agent Name	Created By	Created Date	Last Modified By	Last Modified Date
Support Intelligence Agent	Mithra VishakhaBalmiki Leth...	9/30/2026, 08:44 PM	Mithra VishakhaBalmiki Leth...	9/30/2026, 10:17 PM

Developer Edition

Search...

Developer Edition

Welcome

Report: Support Tickets Intelligence

New Support Tickets Intelligence Report

Enable Field Editing

Search

Filter

Refresh

Edit

3

1

2

Priority Level

Support Ticket Intelligence: Support Ticket Intelligence Name

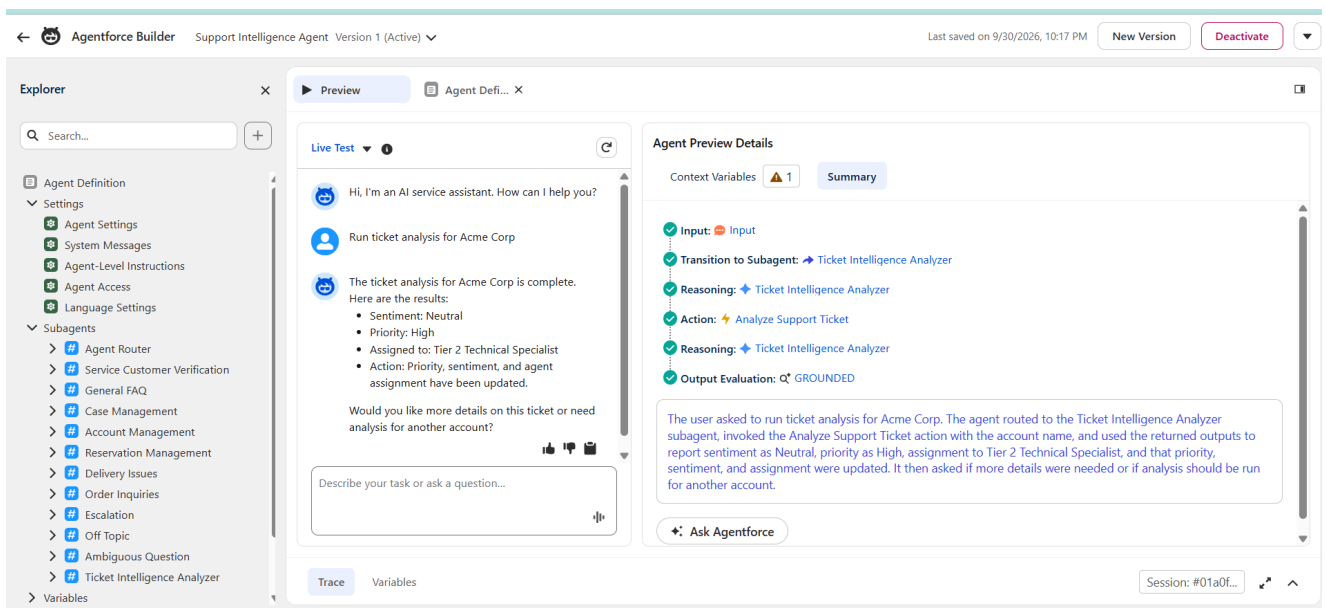
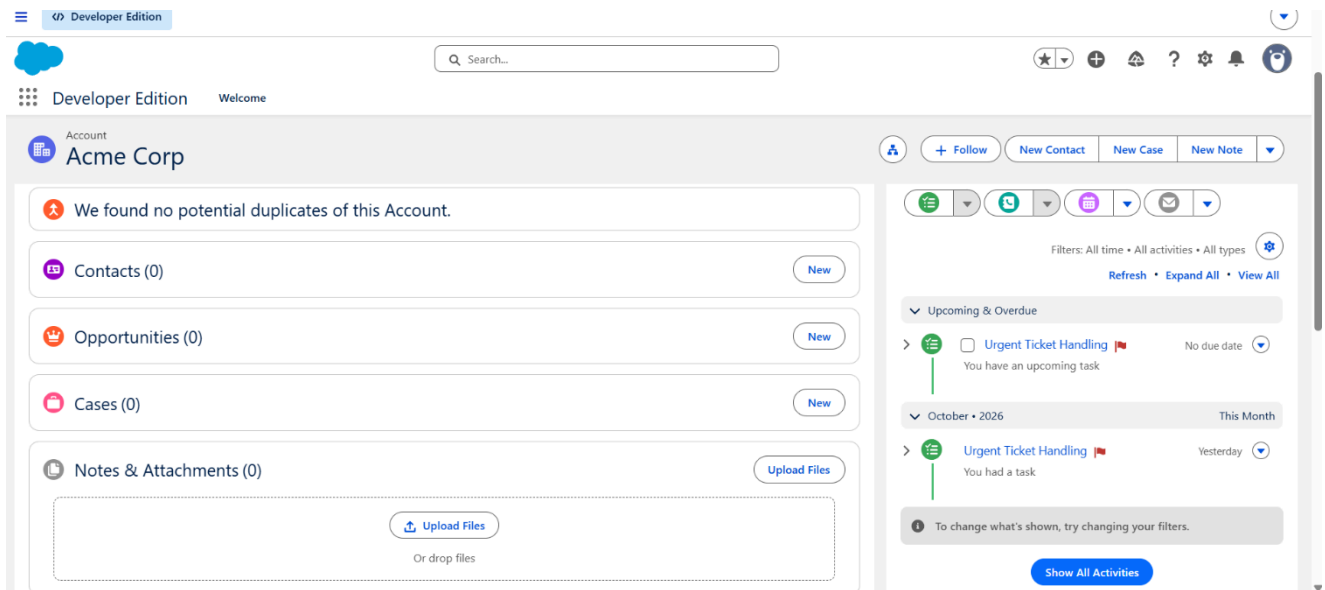
Medium (1)	TKT-1002
Subtotal	
Low (2)	TKT-1001
	TKT-1003
Subtotal	

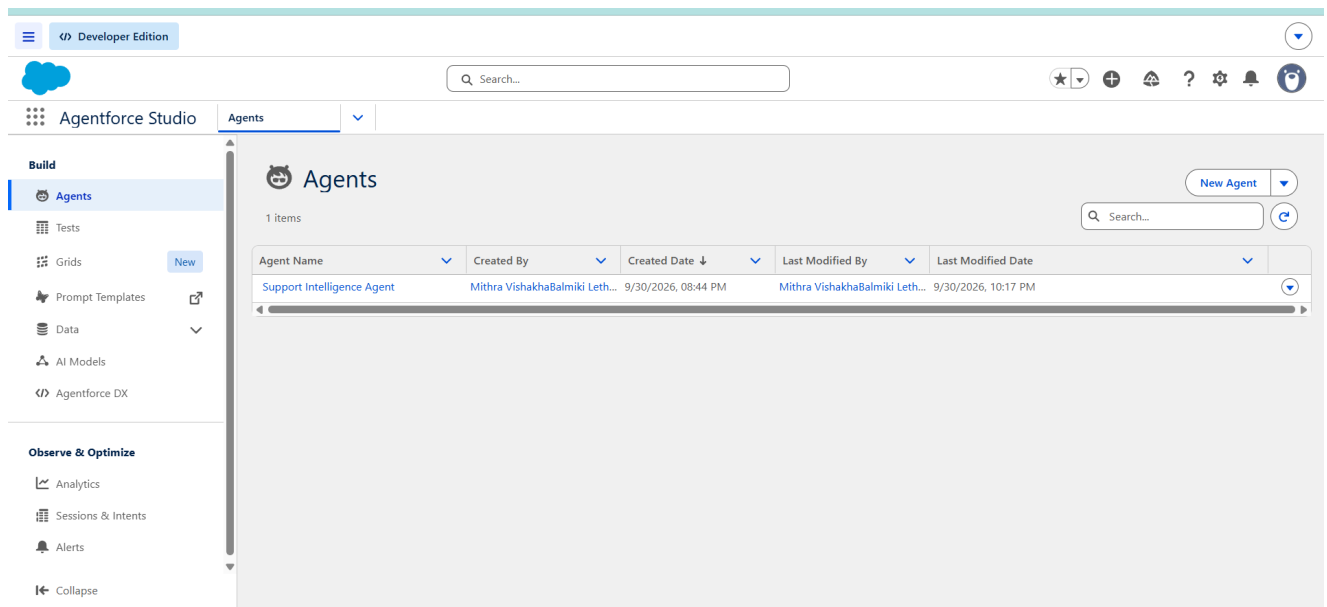
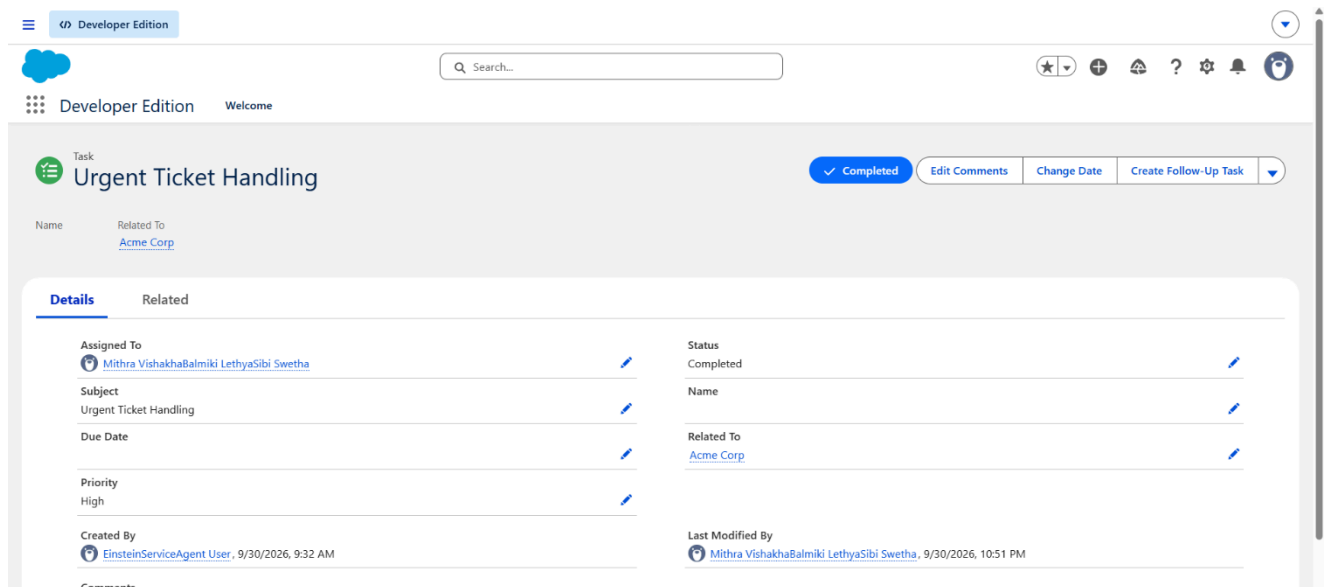
Row Counts

Detail Rows

Subtotals

Grand Total





10. CONCLUSION

The Customer Support Ticket Priority Prediction and Automated Assignment System demonstrates how Salesforce Flow and Agentforce work together to eliminate manual support-ticket handling. The solution retrieves the latest ticket for an Account, analyzes its description, classifies priority, creates an urgent task for High-priority cases, assigns a support level, and returns a clear conversational response. This delivers faster ticket handling, reduced manual effort, intelligent prioritization, and improved team productivity.

11. FUTURE SCOPE

- Add more priority indicators and business rules.
- Extend Agentforce outputs with additional ticket context and resolution history.
- Add more sophisticated SLA monitoring and escalation.
- Introduce analytics for priority trends and resolution performance.

12. APPENDIX

- Appendix A: Custom Object Schema – Support_Ticket_Intelligence__c
- Documents the custom Salesforce object structure, including the fields, data types, and relationships used for support ticket management and analysis.
- Appendix B: Key Salesforce Records – Account, Contact, User, and Task
- Provides the key Salesforce records involved in the solution and shows how customer, support, assignment, and task information are associated.
- Appendix C: Flow Definition – Support_Ticket_Intelligence (Version 8, System Mode)
- Presents the Auto-Launched Flow configuration, including record retrieval, ticket analysis, priority classification, assignment logic, urgent task creation, and SLA risk handling.
- Appendix D: Agentforce Subagent – Ticket Intelligence Analyzer
- Documents the **Support Ticket Priority Analysis** subagent (Support_Ticket_Priority_Analysis), including its purpose, scope, instructions, and interaction with the backend automation.
- Appendix E: Agentforce Action – Flow-Based Support Ticket Intelligence
- Describes the Agentforce action used to invoke the **Support_Ticket_Intelligence** Flow, enabling conversational ticket analysis, priority determination, support assignment, and automated handling actions.
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THANK YOU